

Design Report 1

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EML2322L - Spring 2025

Design Description

This design is for an air-engine-powered food slicer meant to cut cylindrical-shaped food items such as cucumbers or carrots. It uses a combination of a rocker-type four-bar linkage configuration, a dovetail-slot-like 3D-printed chassis slider, a slider cup, a press-fitted stationary blade, and a height adjuster plate to cut different thicknesses of food.

The four-bar linkage is responsible for converting the circular motion of the air engine's flywheel into linear reciprocal motion, repeatedly pushing the slider cup that holds the food back and forth. The linkages are modified off-the-shelf (OTS) components, consisting of flat stainless-steel mounting brackets cut to size and drilled with holes at specified locations. These brackets were selected to reduce cost and manufacturing time, and stainless steel was chosen for its common use as a food-safe material. The floating link is directly connected to the engine, using the air engine's shoulder screw as a pivot point. The coupler link connects to the slider cup, pivoting on a shoulder screw secured with a matching nut. Similarly, the output link pivots on a shoulder screw and nut, with one end attached to the slider chassis. The output link is positioned at the midpoint between the chassis and the flywheel, where all the links converge at a single shoulder screw joint secured with a nut. All joints use M4 shoulder screws. Metric fasteners were chosen due to familiarity with designing linkages in metric units, and since the lab primarily uses imperial tools, the required clearance holes would need to be oversized regardless. To minimize friction and wear, a nylon washer is placed at every joint where links contact each other or a surface.

The slider cup moves along the dovetail-like feature of the 3D-printed chassis, which ensures it remains aligned and constrained to a straight linear path. To facilitate smooth motion, there is a 0.004-0.008-inch free fit tolerance between the chassis and the slider cup. At the opposite end of the chassis, there are four 1/8-inch slots spaced in 0.25-inch increments. These slots accommodate the plate component, allowing for adjustable cutting thickness. A modified OTS blade is press-fitted inside the chassis, positioning its cutting edge 0.25 inches above the surface of the plate component. There is a 0.000-0.002-inch press-fit tolerance between the chassis and the blade. The unmodified OTS blade is a standard, low-cost kitchen butcher knife. Using a modified OTS component instead of a machined blade reduces cost, manufacturing time, and complexity while ensuring the use of stainless steel, a food-safe material. As the most complex component, the chassis was 3D-printed to minimize cost and manufacturing time. Additionally, the elasticity of the plastic allows for secure press-fitting of the blade. The slider cup is CNC-milled from ultra-high-molecular-weight (UHMW) polyethylene, chosen for its lightweight properties, food safety, and low surface friction. Similarly, the plate component is made from UHMW for the same reasons as the slider cup.

Currently, the combination of the air engine and the four-bar linkage enables approximately 3.5 inches of linear displacement for the slider cup. This displacement could be adjusted in future redesigns by modifying the length and position of the four-bar linkage components. However, such changes may introduce stability issues, as they could shift the center

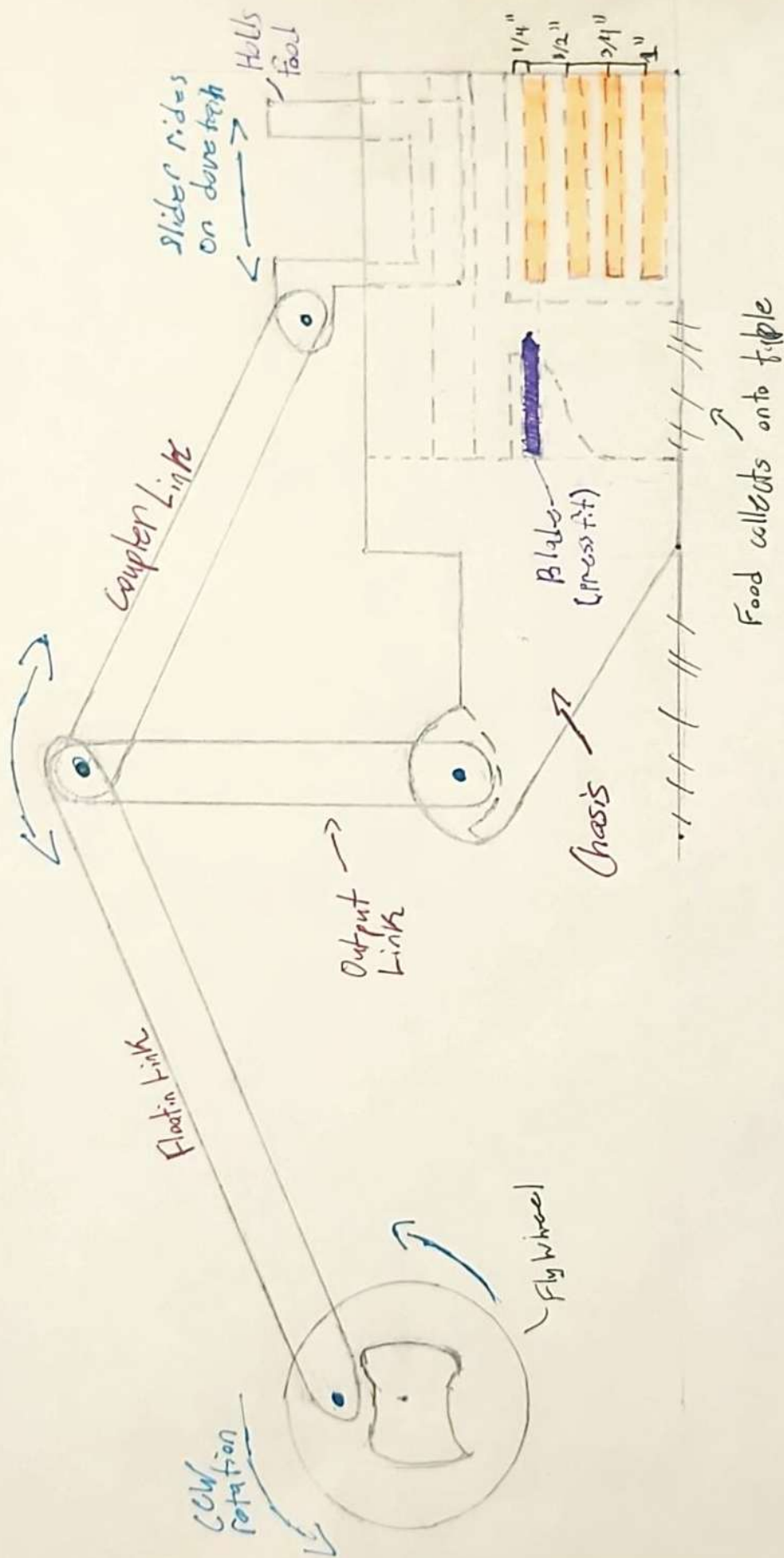
of mass higher above the pivot point. Physics dictates that the farther the center of mass is from the pivot, the more susceptible the system is to tilting. In this case, increasing the linkage length would likely cause the components to tilt against each other, disrupting proper movement. To balance stability and functionality within the given constraints, 3.5 inches was selected as the current displacement.

Conceptual Design Specification Table

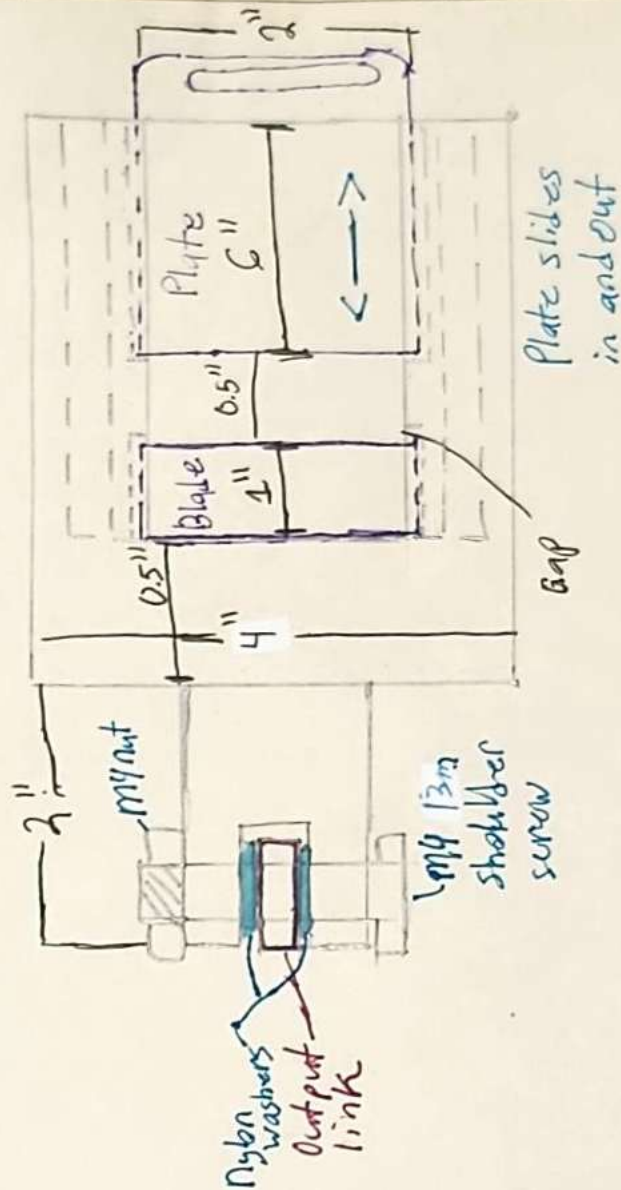
EML2322L Design Specification Table						
Component (Name and/or Part Number)	Description / Size	Critical Surfaces (surface IDs and brief description)	Material	Vendor (if applicable)	Cost	Qty.
Follower Link	4" x 4/5" x 3/25", Flat Straight Braces (Modified)		Stainless Steel	Amazon	\$1.27	2 units per pack each are 12" long; \$7.59/2 = ~\$3.8 per unit; \$3.8/12" ~ 0.32 per inch; 4" * \$0.32 ~ \$1.27
Output Link	2.9" x 4/5" x 3/25", Flat Straight Braces (Modified)		Stainless Steel	Amazon	\$0.92	2 units per pack each are 12" long; \$7.59/2 = ~\$3.8 per unit; \$3.8/12" ~ 0.32 per inch; 2.9" * \$0.32 ~ \$0.92
Coupler Link	2.3" x 4/5" x 3/25", Flat Straight Braces (Modified)		Stainless Steel	Amazon	\$0.73	2 units per pack each are 12" long; \$7.59/2 = ~\$3.8 per unit; \$3.8/12" ~ 0.32 per inch; 2.3" * \$0.32 ~ \$0.73
Blade	2" x 1" 0.1" (Modified)	Modified OTS, cut-to-size, press-fit into chassis	Stainless Steel	Amazon	\$8.99	1.00 Stated Price of OTS Component
Chassis	Chassis Volume Est: 8" * 4" * 2.8" = 89.6 in^3	Slicer main body; Has dovetail channel, thickness plate slots, press fit mount for blade	PETG Filament	Amazon	\$7.70	\$22.9/kg; Printer Settings: 0.24mm Layer Height, 2 walls, 15% Gyroid Infill; Chassis Volume Est: 8" * 4" * 2.8" = 89.6 in^3; Slicer Est Cost: \$7.70
Slider Cup (8702K545)	Slider Cup Volume Est: 3" x 3" x 2" = 18 in^3	Slides in Chassis's dovetail channel, holds food for cutting	Slippery UHMW Polyethylene	McMaster Carr	\$8.46	1.00 Stock Size: 1" x 3" x 1" (L x W x H); \$50.75/ft = ~\$4.23/in; 2" * \$4.23 = ~\$8.46
Plate (8702K27)	Plate Volume Est: 6" x 3" x 0.125" = 2.25 in^3	Acts a thickness adjuster, interacts with plate slots on Chassis	Slippery UHMW Polyethylene	McMaster Carr	\$2.03	1.00 Stock Size: 1" x 3" x 1/8" (L x W x H); 4.05/ft = ~\$0.34/in; 6" * \$0.34 = ~\$2.03
Metric 316 Stainless Steel Shoulder Screws (90269A222)	Ø5mm Shoulder x 13mm L x M4 Thread x 5mm TL	Acts a pivot point for linkages	317 Stainless Steel	McMaster Carr	\$4.10	3.00 5 units per pack; \$8.2/4 = \$2.05 per unit; \$2.05 * 2 = \$4.10
Metric 18-8 Stainless Steel Shoulder Screws (90265A127)	Ø5mm Shoulder x 6mm L x M4 Thread x 5mm TL	Acts a pivot point for linkages	18-8 Stainless Steel	McMaster Carr	\$1.68	1.00 5 units per pack; \$6.72/4 = \$1.68 per unit
18-8 Stainless Steel Nylon-Insert Locknut (93625A150)	Ø7mm Head x 5mm H x M4 x 0.7mm Thread	Keep shoulder screws in place	18-8 Stainless Steel	McMaster Carr	\$0.21	3.00 100 units per pack; \$7.03/100 = ~\$0.07 per unit; \$0.07 * 3 = ~\$0.21
Plastic Washers (95610A360)	M5 x Ø5.3mm ID x Ø9mm OD x 1.2mm Thk	Reduce wear; Located at every joint/surface contact point	Nylon	McMaster Carr	\$0.54	6.00 100 units per pack; \$8.84/100 = ~\$0.09 per unit; \$0.09 * 6 = ~\$0.54
Total =					\$36.63	

Concept Sketches

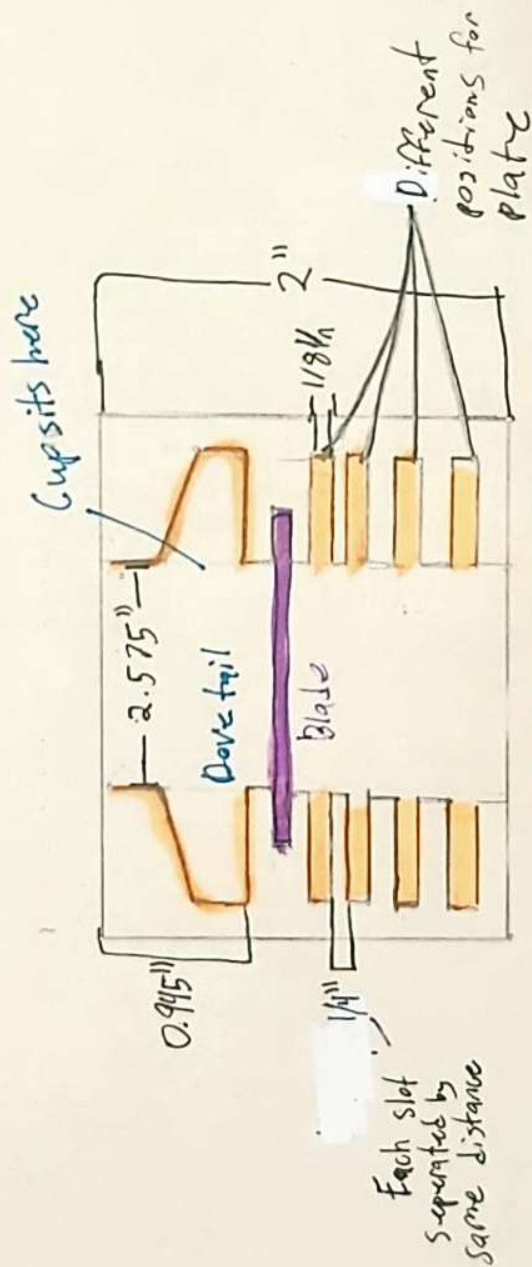
Front View-Full



Chassis Top View (4 bar linkage & Slider Cup) not included



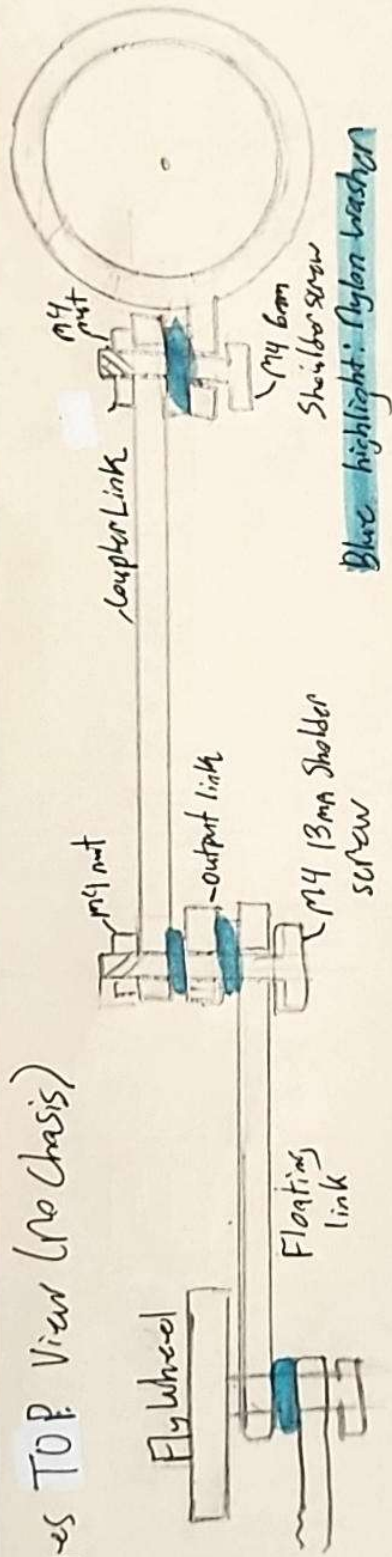
Chasis Right Sid View



orange highlight: 0.004" - 0.008" Free fit tolerance

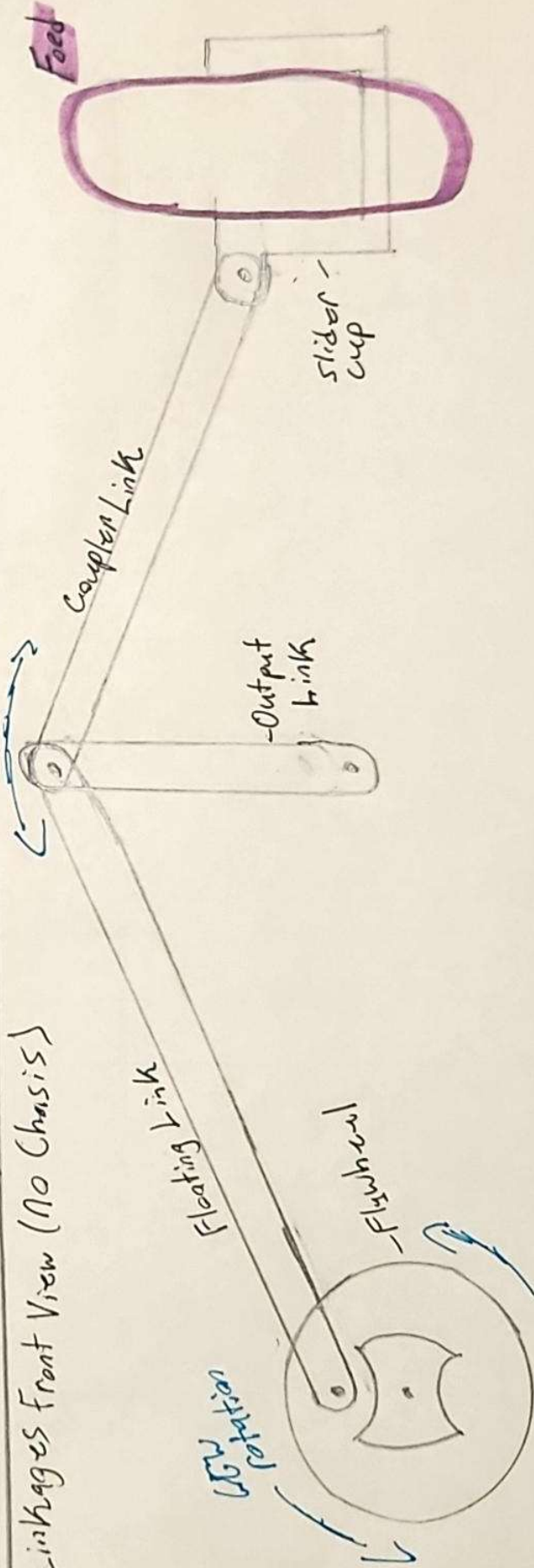
purple highlight: 0.00" - 0.002" Press fit tolerance

Linkages TOP View (No Chassis)



Linkages have 3/25" thickness

Linkages Front View (No Chassis)



Chassis front Section View

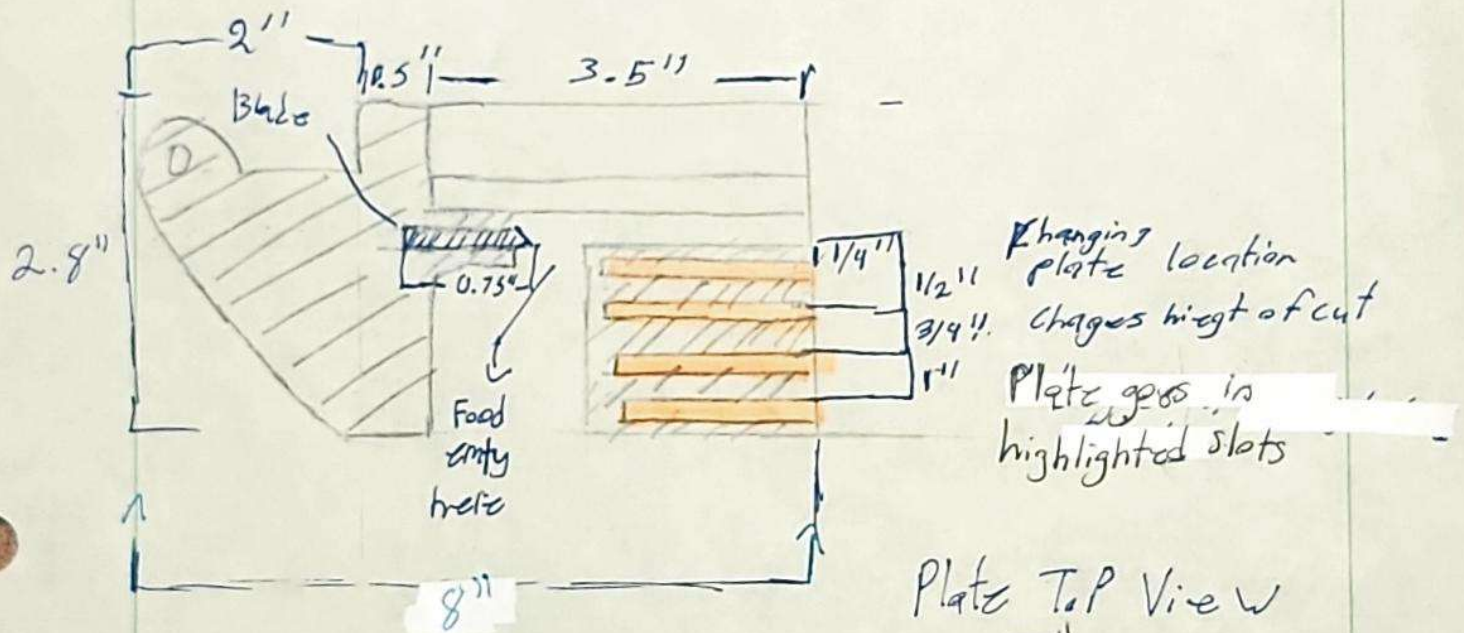
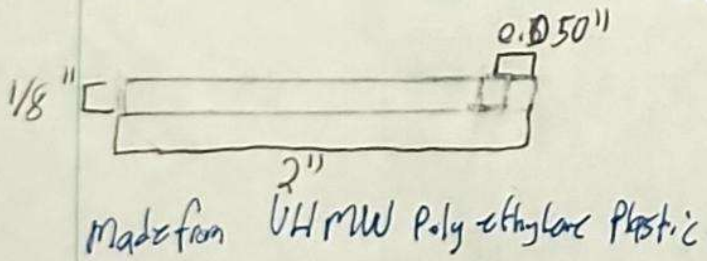
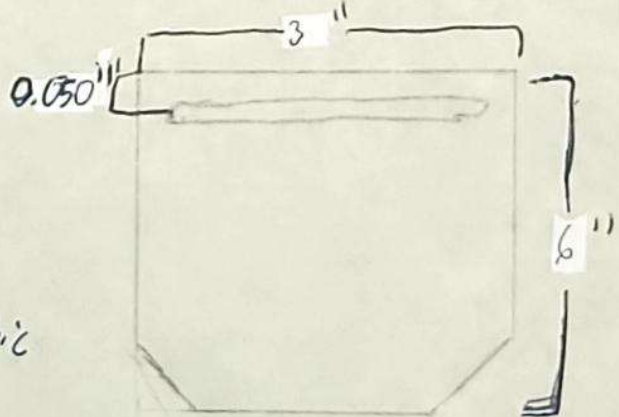


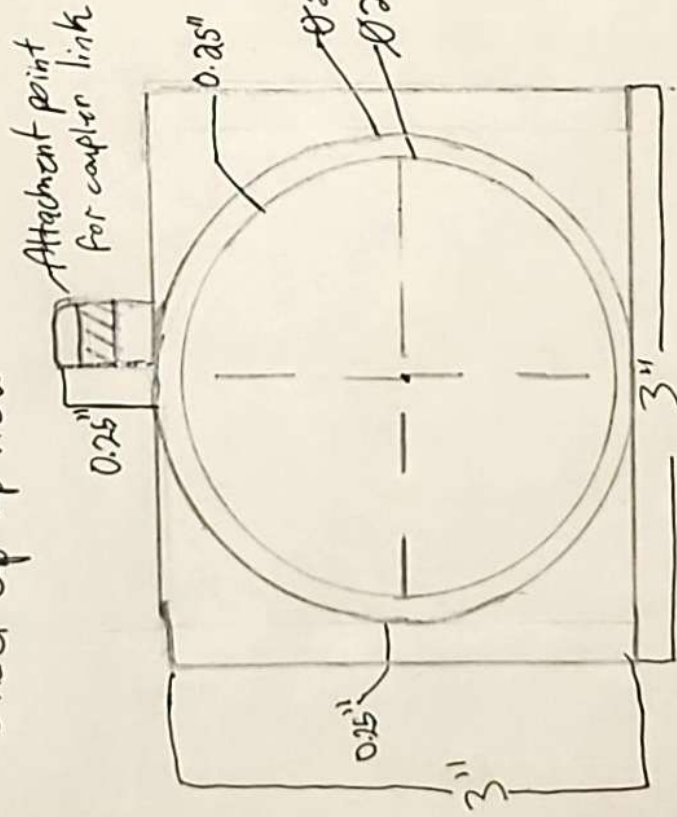
Plate Front View



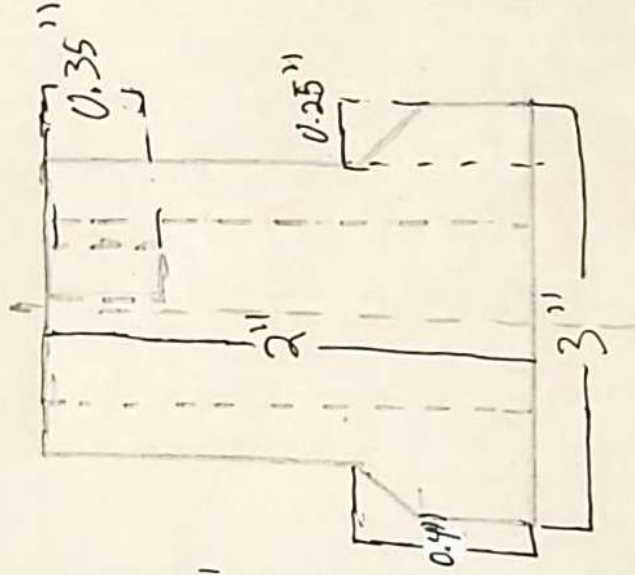
Platz T.P View



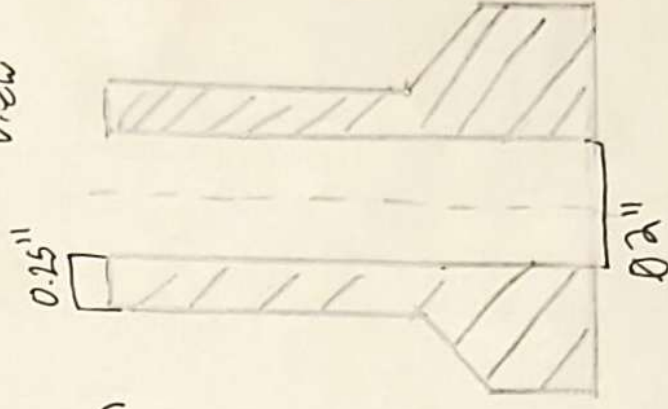
Slider Cup Top View



Slider Cup Right View



Slider Cup Right Section Cut View



CNC from Slippery UHMW Polyethylene

OTS Knife Top View



-This is Base Dimensions of Knife that will be modified by cutting

-It will be press fitted into 3D printed part

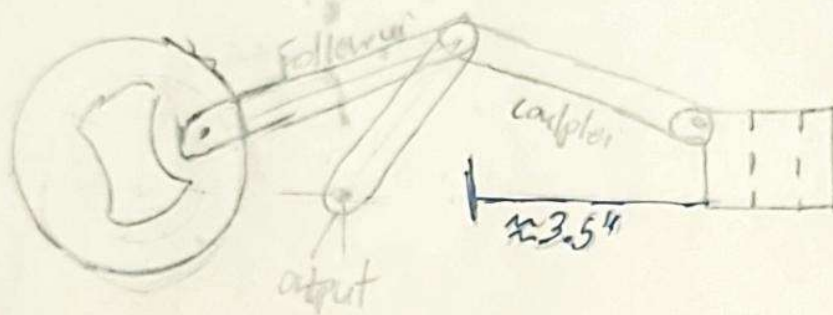
Prod Section is cut from original knife and used in Slicer

OTS Knife Top View (Cut)



Four Bar Linkages Mechanism front view (1/2 scale)

Starting position



End position

Linear
Displacement

